

constant, net force and acceleration decreases (to zero) [A1].

2 (a)

When the gas is heated, the molecules **gain kinetic energy**. As the molecules move faster, they also **collide with the piston more frequently**. [B1]

The **change of momentum** experienced by the molecules upon collision with the wall, causes an impact **force on the piston wall pushing it** to the right thereby doing work in causing the gas to **expand**. [B1]

Credit was given for those who gave  $W = P\Delta V$  but they need to explain how the pressure originate from the collision of the gas particles on the wall and that the volume increases.

2 (b)(i)

Any two possible answers [B1x2]

1. There can be work done against friction due to the piston rubbing against the wall which dissipates the supplied energy.
2. The heat generated does not totally transfer into mechanical work done as some is given to raise the internal energy of the gas.

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3. *The wall of the vessel is not a perfect insulator and it might conduct some of the heat away from the*